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Seasonal distribution of chlorophyll on mudflats in New South Wales, Australia measured by field spectrometry and PAM fluorometry

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ABSTRACT

Variability of chlorophyll (as an index of micro-algal abundance) between warm and cool seasons at different heights on (distances across) the shore was investigated on intertidal mudflats in warm-temperate Australia. Chlorophyll was measured using ratios of reflectances from field spectrometry and minimal fluorescence (F_0) from PAM fluorometry to compare patterns obtained using these two methods. A single sampling period comprised 2 days of sampling, one for each mudflat, with 2 sampling periods nested within each month, 2 months within each of a cool and warm season in each of 2 years. Large differences in amounts of chlorophyll were found between the two mudflats, although spatial and temporal patterns of variation were generally similar. There were greater amounts of chlorophyll in the cooler months than in the warmer months in each location in each year, which contrasts with many of the patterns reported from elsewhere. There was more chlorophyll on the upper than on the lower shore and the increases from summer to winter were generally greater at the higher levels. Large variation in chlorophyll from week to week within each month demonstrated the need for adequate replication in studies of seasonal patterns of variability. Measurements made by a field spectrometer and a PAM fluorometer were largely consistent, but, at certain times, they showed an opposite pattern. The reasons for these differences were investigated further by looking at differences in other pigments, but the different results from the two methods could not to be explained by changes in composition of the micro-algal assemblage and, as yet, remain unexplained.

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1. Introduction

Benthic micro-algae or microphytobenthos (MPB) are important in the ecology of soft-sedimentary habitats (reviewed by Admiraal, 1984; Miller et al., 1996). They include diatoms, euglenids and cyanobacteria and the juvenile stages of macro-algae (Pinckney and Zingmark, 1993). MPB make a significant contribution to primary productivity (Cadee and Hegeman, 1974; MacIntyre et al., 1996; Underwood and Kromkamp, 1999), which is regulated by a number of inter-related factors, including sediment-type, temperature and amounts of incident sunlight (Shaffer and Onuf, 1983). MPB are a primary food-source for some fish (Almeida et al., 1993; Yang et al., 2003), numerous meiobenthic and macrobenthic organisms (Carman and Thistle, 1985; Newell et al., 1995) and, possibly, birds (Elner et al., 2005). MPB which have been resuspended into the water column may also be food for pelagic grazers (de Jong and van Beusekom, 1992).

Spatial distributions of MPB have been linked to patterns in some fauna (Montagna et al., 1983; Decho and Castenholz, 1986; Decho and Fleeger, 1988), although many studies have shown a lack of correlation between algae and fauna (e.g. Barnes and de Villiers, 2000; Lund-Hansen et al., 2002; Tolhurst and Chapman, 2007) and the factors contributing to this relationship are complex. For example, grazers may respond positively to MPB, but grazing can also influence variability in MPB (Buffan-Dubau and Carman, 2000) and excretion by grazers may increase local amounts of MPB (Connor et al., 1982). MPB also have an important role in the stabilisation of soft sediments through secretion of extracellular polymeric substances (e.g. Paterson, 1989; Austen et al., 1999; Riethmuller et al., 2000; Tolhurst et al., 2002).

Amounts of chlorophyll, often used as an index of biomass of MPB, are very variable in space, both horizontally (Pinckney and Sandulli, 1990; Buffan-Dubau and Carman, 2000; Murphy et al., 2008b) and vertically (Joint et al., 1982; de Jonge and Colijn, 1994), or among different microhabitats at the same tidal height (Tolhurst and Chapman, 2007). Therefore, sampling must be adequately replicated to estimate amounts of chlorophyll at larger spatial scales (Migne

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et al., 2004). Many studies have reported more chlorophyll on the upper than the lower parts of intertidal mud- or sandflats (Davis and McIntire, 1983; Underwood and Paterson, 1993; Underwood, 1994; Brotas et al., 1995; de Brouwer et al., 2000; Paterson et al., 2000; Staats et al., 2001; Thornton et al., 2002, but see Riznyk and Phinney, 1972). The reasons for this pattern are not known, but it has been attributed to disturbance of lowshore sediments by re-suspension (de Jong and de Jonge, 1995; Staats et al., 2001).

MPB also vary temporally at many different scales. Over short periods, vertical migration of MPB through the sediment can occur in response to tidal emersion (Consalvey et al., 2004; Honeywill et al., 2006; Jesus et al., 2006), although there can be large day-to-day variability irrespective of tidal condition (Tolhurst and Chapman, 2005). Variability over longer timescales (e.g. seasonal or annual changes) has been investigated on several mudflats, predominantly in Northern Europe (reviewed by Underwood and Kromkamp, 1999). There may be more chlorophyll in summer than in winter (e.g. de Jong and de Jonge, 1995; Guarini et al., 1998; Staats et al., 2001; Migne et al., 2004), peaks in spring and autumn, or little or no seasonal variability (e.g. Brotas et al., 1995; Rossi et al., 2001; Thornton et al., 2002). Seasonal patterns may also differ between locations in the same estuary (Wolfstein et al., 2000), for only some positions along a transect (Davis and McIntire, 1983; Staats et al., 2001), or for only some years (Rossi et al., 2001). Lack of seasonal patterns, for example, in the Ria de Arosa in Spain (Varela and Penas, 1985) or the Tagus estuary in Portugal (Brotas et al., 1995), may be due to milder winters at these latitudes. Rapid increases in chlorophyll may also occur at any time of the year in response to rapidly changing environmental conditions, thus masking any seasonal patterns (Underwood and Kromkamp, 1999).

Although there have been numerous studies on seasonal and vertical patterns of micro-algal abundance on mudflats in Europe, there are few comparable data for mudflats in New South Wales, Australia, where conditions are quite different. Seasonal changes in climate are relatively mild. Diatoms and euglenids, which are often a major component of the MPB in European estuaries (Admiraal, 1984; MacIntyre et al., 1996; Miller et al., 1996; Paterson et al., 1998), are less common in NSW, where fine filaments of green algae can also be intimately dispersed among the surface grains of sediment (Inglis, 1996). The intertidal shores in many NSW estuaries are also much less extensive than on many of the mudflats in Europe, where vertical patterns of chlorophyll have been measured over ranges of 100s of metres (Staats et al., 2001) or even kilometres (Paterson et al., 2000). The tidal range is also only 2 m, less than that found in many European estuaries.

Many previous studies have been inadequately replicated to determine any generality of the seasonal or vertical patterns of chlorophyll reported. For example, Paterson et al. (2000) sampled at 4 intertidal heights in a single bay, but at only one station per height, thus potentially confounding vertical patterns of difference with spatial variation at any particular height. Similarly, studies where 3-monthly periods have been sampled without any temporal replication within each season or over multiple years (Staats et al., 2001) are often used to infer seasonal patterns when these cannot be distinguished from random or other variation occurring at smaller temporal scales (Underwood, 1994).

Here, chlorophyll was measured using field spectrometry (Carrere et al., 2004; Murphy et al., 2005a; Kromkamp et al., 2006) and Pulse Amplitude Modulated Fluorometry (PAM; Honeywill et al., 2002; Jesus et al., 2005; Serodio et al., 2006) at temporal scales of weeks, months, seasons and years. These data thus have the levels of replication necessary to distinguish seasonal patterns of variation in chlorophyll from smaller (weekly, monthly) or larger (annual) temporal scales of variability. In addition, because samples were taken at different distances from the shore in replicate sites in each of two mudflats in Sydney Harbour, they also measure the consistency of such patterns among different sites on a shore, among shores and at different distances from shoreward to seaward. Because these mudflats have very little slope and are constrained landward by mangrove forests or seawalls, there was little vertical extent. Nevertheless, the period of emersion varied more than 2-fold between the seaward and shoreward sites. Field spectra and the F_0 measured by PAM should, theoretically, give similar results because they both measure chlorophyll, although other studies have shown different patterns of variation in chlorophyll according to the method used to sample it (Paterson et al., 2000). The relative abundances of other pigments were determined from the field spectra to test the hypothesis that these differences were due to changes in composition of the algal assemblage.

2. Materials and methods

2.1. Field sites

Two small (>1 km in length) embayments, Tambourine Bay (33° 49′ 42.89″ S, 151° 09′ 47.55″ E) and Brays Bay (33° 50′ 01.34″ S, 151° 05′ 28.09″ E) in the upper reaches of Sydney Harbour were studied. The upper shores of both bays were bordered by mangroves, which varied from about 29 m wide at Tambourine Bay to 38 m at Brays Bay. The mud under the mangrove canopy and amongst the pneumatophores, which may extend beyond the canopy, are different habitats from the open mudflats (Chapman and Tolhurst, 2004); only the open areas of mud, without mangrove canopy or pneumatophores were sampled here. Sediments in each bay are comprised of variable amounts of sand, mud and silt. Algal assemblages are dominated periodically by fine green filamentous algae. Average maximal air temperatures, obtained from a local station by the Australian Bureau of Meteorology, were between 19.5 °C and 23.3 °C in the cool season and 24.4 °C and 27.4 °C in the warm season.

2.2. Sampling strategy

Sampling was from November 2005 to September 2007. Two seasons were sampled: a warm season (centred on Austral summertime) and a cool season (centred on Austral wintertime). To test hypotheses about seasonal differences with adequate temporal replication to unconfound seasonal and other temporal differences (e.g. review by Underwood, 1994), in each of 2 years, sampling was done in each of 2 months, randomly chosen in each season and in 2 weekly sampling periods in each month. A single sampling period comprised 2 days of sampling, 1 for each bay. Where possible, bays were sampled on consecutive days, but never more than 3 days apart. Thus, there were 2 sampling periods, separated by 2 weeks, within a month. This was replicated 4–6 weeks later in each season.

All sampling was done during low tide. In each bay, data were collected from 3 replicate plots of 1 m^2 about 30 m apart at each 4 m distance from the lower border of the mangroves down to the level of low tide. On subsequent occasions, different plots at the same distances were sampled, to maximize independence; plots were therefore nested in the combinations of all other factors (weeks, months, seasons; bays, and distances). The distances sampled were at different heights as measured by their period of emersion during low tide (Table 1). We chose this method for sampling because there was little vertical range, the sediment moves and small-scale differences in height across the shore can make it difficult to find exactly the same height at subsequent times. Thus, henceforth, instead of different heights we sampled at regular distances downshore. The maximal range of distances sampled was 16 m. The smaller tidal range in NSW estuaries and the mangrove occupying the upper part of the mudflat restricted

Table 1
Average relative proportions of time each distance was emersed out of the total period of emersion during a typical spring low tide.

		Brays Bay	Tambourine Bay
Upper shore	Distance 1 (0 m)	0.30	0.31
	Distance 2 (4 m)	0.26	0.21
	Distance 3 (8 m)	0.21	0.16
	Distance 4 (12 m)	0.17	0.11
Lower shore	Distance 5 (16 m)	0.12	0.08

the range of distances (and therefore heights) that could be sampled (Table 1).

2.3. Field spectrometry

Reflectance spectra (350–1050 nm) were recorded using a field spectrometer (FieldSpec Pro, Analytical Spectral Devices, Boulder, Colorado). Four replicate spectra were obtained from the mud surface at haphazardly located positions within each plot. Areas covered by water were avoided when sampling. Prior to each spectrum being recorded, a calibration spectrum was recorded from a ~99% reflective panel (Spectralon, Labsphere, North Sutton, New Hampshire). Spectra were collected using an 8° fore-optic, from a height of 35 cm; thus each spectrum measured an area of mud 18.7 cm^2. Each replicate spectrum was an average of 30 individual spectra.

In the laboratory, reflectance spectra were obtained by dividing each mud spectrum by its corresponding calibration spectrum. Chlorophyll absorbs strongly in the red part of the spectrum, but algal cells scatter light in the near-infrared (NIR). The amount of absorption by chlorophyll-*a* can be estimated to within 2.37 $\mu g\ cm^2$ of conventional sampling methods, using a simple ratio of reflectances at 750 nm (R750; where chlorophyll does not absorb) and 672 nm (R672; where chlorophyll is maximally absorptive); for details, see Murphy et al. (2005a). The ratio value (R750/R672), termed the spectrometer ratio, increases with increasing amounts of absorption by chlorophyll-*a*. Where chlorophyll values were required, the ratio was converted to chlorophyll using the following equation, with data expressed per unit area in accordance with Murphy et al. (2005b):

$$\text{Chlorophyll}\left(\mu g\ cm^{-2}\right) = 5.17^{*}(R750/R672) - 2.51 \qquad (1)$$

To test the hypothesis that changes in the algal assemblage had occurred over specific periods of time, absorption by other pigments was measured. A change in the relative amount of pigments may indicate a change in the composition of the algal assemblage (Millie et al., 2002). Amount of absorption by pigments (including chlorophyll) was determined using derivative analysis of reflectance spectra. Ideally, 4th-derivative spectra should be used to quantify pigments (Bidigare et al., 1989), but, because of the amount of noise in our spectra, 2nd-derivatives were used as an alternative. 2nd-derivative spectra have peaks at wavelengths where there is absorption by pigments. The height of this peak is indicative of the amount of absorption and, hence, amount of pigment. These were calculated with a 30 nm smoothing interval using the method of Savitzky and Golay (1964) and the amount of each pigment in each spectrum was calculated as the maximal derivative value above the zero-baseline. Eight absorptions (termed pigment bands) were identified in the spectra and the identity of their respective pigments was inferred from published absorption maximima of pigments *in vivo*: 509 nm (carotenoids/xanthophylls), 544 nm (carotenoids/xanthophylls), 552 nm (carotenoids/xanthophylls), 588 nm (chlorophyll-*c*), 616 nm (chlorophyll-*a*), 636 nm (chlorophyll-*c*/phycocyanin), 665 nm (chlorophyll-*a*), and 686 nm (chlorophyll-*a*).

2.4. PAM fluorometry

A PAM fluorometer (Diving PAM/B; Heinz Walz GmbH, Effeltrich, Germany) was used to measure minimal fluorescence, F_0 (Honeywill et al., 2002; Consalvey et al., 2005). Four replicate PAM measurements were recorded from haphazardly located positions in each plot. PAM measurements were not spatially matched with reflectance spectra. Prior to measurement, areas of mud were dark-adapted for 15 min, by placing an upturned crucible, which had been lined with tin-foil (to ensure complete darkness), on the mud surface. To enable comparison across measurements, the settings of the PAM were kept the same across all sampling times. The diameter of the fibre-optic was 0.6 cm, so each measurement was 0.28 cm^2. A plastic spacer was used to ensure that all measurements were made at a constant distance of 2 mm from the surface of the mud.

2.5. Analyses of data

The general structure of analyses is summarised in Table 2 where temporal scales are Years (2 levels, random) and Season (2 levels, warm versus cool seasons, fixed), months (2 months

Table 2
Analyses of data from spectrometer (ratio data) and PAM (F_0) for each Bay (see text for details). Years were random (2 levels); Seasons, fixed (2 levels); Distances, fixed (3 levels); Months in each year and season, random (2 levels); Weeks in each month, random (2 levels); Plots in each combination of all after factors, random (2 levels); 4 replicates were sampled in each plot at each time. Sources of variation not significant at $P = 0.25$ were eliminated (^{e}) terms used as divisors in tests are indicated as superscripts on *F*-values; main effects involved in significant interactions are not.

Source of variation		df	Brays Bay						Tambourine Bay					
			(a) Ratio			(b) $F_0 \times 10^{-4}$			(c) Ratio			(d) $F_0 \times 10^{-4}$		
			MS	*F*	*P*	MS	*F*	*P*	MS	*F*	*P*	MS	*F*	*P*
1	Years = Y	1	0.8	0.1^{10}	>0.75	2365	1.7^{10}	>0.70	0.03	0.4^{10}	>0.50	319	3.3^{10}	>0.10
2	Seasons = S	1	57.1	7.2^{10}	<0.03	7025			0.23	3.7^{10}	>0.05	549	2.1^{4}	>0.35
3	Distances = D	2	1.4	2.6^{11}	>0.10	184			0.02	2.1^{12}	>0.10	3	1.0^{12}	>0.35
4	Y × S	1	4.8^{e}	0.6^{10}	>0.45	198^{e}	0.1^{10}	>0.70	0.06^{e}	1.0^{10}	>0.35	266	2.8^{10}	>0.10
5	Y × D	2	0.5^{e}	0.9^{11}	>0.40	90	1.6^{11}	>0.20	0.01^{e}	1.3^{12}	>0.25	3	1.1^{12}	>0.35
6	S × D	2	1.2	2.2^{11}	>0.10	343	6.2^{11}	<0.01	0.003	0.4^{12}	>0.65	0.03	0.01^{12}	>0.95
7	Y × S × D	2	0.7	1.9^{11}	>0.18	128	2.3^{11}	>0.10	0.001	0.1^{12}	>0.90	0.7	0.2^{12}	>0.80
8	Months (Y × S) = M (YS)	4	9.7^{e}	1.2^{10}	>0.35	383^{e}	0.3^{10}	>0.85	0.04^{e}	0.7^{10}	>0.60	18^{e}	0.2^{10}	>0.90
9	D × M (YS)	8	0.4^{e}	0.7^{11}	>0.70	51^{e}	0.9^{11}	>0.50	0.005^{e}	0.6^{12}	>0.75	1^{e}	0.5^{12}	>0.85
10	Weeks (M (YS) = W (M (YS))	8	7.9	27.2^{13}	<0.0001	141	46.5^{12}	<0.0001	0.06	8.0^{12}	<0.0001	95	31.5^{12}	<0.0001
11	D × W (M (YS))	16	0.5	1.9^{13}	<0.05	56	1.8^{12}	>0.05	0.004	0.6^{12}	>0.85	0.8	0.3^{12}	>0.95
12	Plots (DW (M (YS))	48	0.3^{e}	1.1^{13}	>0.25	30	1.4^{13}	<0.05	0.008	3.7^{13}	<0.0001	3	3.6^{13}	>0.0001
13	Residual	288	0.3			21	–		0.002	–		0.8	–	

randomly chosen and nested in each combination of Year and Season) and Weeks (1 day sampled in each of 2 weeks randomly chosen in each month, i.e. nested in month, year and season). The spatial scales were Bays (randomly chosen) and Distances (evenly spaced distances, 5 levels, fixed).

Loss of some data, due to variations in the area of mudflat emersed at some sampling times, restricted the full analysis. Data were available for 2 plots at all distances in both Bays, although not all weeks and months could be sampled at the lower 2 distances. There were, however, at least 2 periods of sampling scattered across the 2 months in each season (i.e. not both weeks in both months, but 2 of the 4 weekly times of sampling were available in every season and year) for 2 plots. Thus, data were analysed for the full nested set of times (Seasons, Months, and Weeks) for the upper 3 distances and, separately, for 2 sampling periods (2 levels, random) in each year and season for all 5 distances. Where more than 2 periods were available, 2 were picked at random for analyses.

Valid *F*-ratio tests were constructed by eliminating higher-order interactions provided that they had $P > 0.25$ for their *F*-ratios (Winer et al., 1991). The remaining sources of variation were then recalculated (Fletcher and Underwood, 2002). New tests were then constructed from first principles of the design (see Underwood, 1997 for a complete explanation) and the process continued until all required tests were completed. The sources of variation that were eliminated and those used as divisors for tests are identified with analyses (Table 2). Preliminary analyses, which included Bays as a factor, showed that there were significant interactions between Seasons and Bays and between Bays and Months. Because of these issues and the greater amount of chlorophyll found at Brays Bay (see Results), the bays were analysed separately.

3. Results

The major finding was a clear seasonal pattern in each Bay, with cool months having larger amounts of chlorophyll than did warm months (Table 3). At Brays Bay, for measures of chlorophyll using the spectrometer ratio, there were significantly greater values in the cooler than the warmer season (Table 2a, Seasons) at all distances across the shore (i.e. no significant interaction with Distance). Thus, the differences between cooler and warmer seasons shown in Fig. 1 were of similar magnitudes for all distances. At each distance, there was a significant negative correlation of amount of chlorophyll with temperature ($r = -0.56$ to -0.70, all $P < 0.05$, 14 df). Variability in chlorophyll values between sampling times was much greater in the cool than in the warm season (note S.E.s for the 2 periods for different distances in Table 3). Brays Bay generally had more chlorophyll than Tambourine Bay (Figs. 1, 2; Table 3). Mean values of chlorophyll in Tambourine Bay, even in the cooler season, were always around those for the warmer season in Brays Bay (Fig. 1a).

There was very great variation from week to week in all 4 analyses (F_0 and the spectrometer ratio for each Bay; Table 2). At Brays Bay, this was not the same at the different distances (a significant D × W(M (YS)) term in Table 2a). In general, however, weekly differences were significant for both variables in each bay and constituted a very large source of variation (note the sizes of mean squares for this term in analyses in Table 2). Not surprisingly, therefore, there were no significant differences between months or years in the analysis. The variability among months and that for interactions between distances and months were eliminated from all analyses (terms 8 and 9 in analyses in Table 2) because these components of variation were very small and there was no evidence for them being non-zero.

The other large source of variation in the data was among plots at spatial scales of about 30 m, which was significant for F_0 in each bay and for the spectrometer ratio from Tambourine Bay (analyses b, c, d, in Table 2). This is illustrated for one set of data (F_0) for one distance at Brays Bay (Fig. 2b). The measures of F_0 were, other than among plots and weeks, only significantly variable for the Season × Distance interaction (Table 2a). This is illustrated in Fig. 2(a–d) and for the 3 distances analysed in Table 3. There were greater F_0 values in the cooler than the warmer periods for each distance and the difference between seasons was smaller further downshore (Distance 3 < Distance 2 < Distance 1). The mean values in the warm season were much more similar at the 3 distances than in the cooler period, but differences among distances were not significant.

Given the "noise" found for differences among weeks and plots (see above), the analyses of data for Tambourine Bay showed no significant differences between other sources of variation. Nevertheless, the patterns of differences in mean values through time were consistent at the distances sampled (e–h in Figs. 1 and 2). They were also very consistent between the 2 bays (compare a–d with e–h in each figure). This was also shown for the mean amount of chlorophyll in the 2 seasons at 3 levels (see Table 3). This was the same pattern that was significant at Brays Bay – similar amounts of chlorophyll at the 3 distances in the warmer season, more chlorophyll overall in the cooler season and the seasonal difference decreasing with increasing distance from the top of the shore (Table 3).

The spectrometer ratio and F_0 followed a similar pattern across sampling times. At some times of sampling, however, the 2 measures showed different patterns. For example, the spectrometer ratio for Brays Bay (Fig. 1a) at sampling times 5, 6 and 7 (12/08/06, 24/08/06 and 21/09/06, respectively) showed a general decrease, but F_0 showed an increase. This difference between the 2 methods of sampling may have been related to a change in the composition of the assemblage of MPB, which might have caused the relative amounts of pigments to change relative to the amount of chlorophyll-*a* present. To test this hypothesis, a multivariate analysis (ANOSIM, Primer; Clarke, 1993) was done on the pigment bands extracted from the 2nd-derivative spectra. No change in the relative amounts of the pigments was detected (at $P = 0.05$), suggesting that no change in the composition of the MPB assemblage had occurred.

Although not presented, the analyses across all 5 distances revealed essentially the same findings. There was more chlorophyll in the cooler than the warmer season (significantly so for Brays Bay for the spectrometer ratio), with the difference varying with distance from the top of the shore for the F_0 measures. The patterns through time for distances 4 and 5 (i.e. lower on the shore) were virtually identical (Distance 4 is also illustrated in Figs. 1 and 2). Similar changes in chlorophyll were found between seasons in each year, as indicated by the non-significant Year × Season interactions measured by the spectrometer ratio in each bay and by F_0 at Brays Bay. At Tambourine Bay, F_0 increased from warmer to cooler months in each year, but there was a significant Year × Season interaction. In the warmer months, mean F_0 over all heights and

Table 3
Amounts of chlorophyll ($\mu g\,cm^{-2}$) at different distances in Brays Bay and Tambourine Bay in cooler and warmer periods of the year. Data are means (S.E.; $n = 64$) from 2 weeks in each of 2 months in each of 2 years, for 2 replicates from each of 2 plots at each distance. Data are converted from ratios using the regression described in the text.

	Brays Bay			Tambourine Bay		
	Cool	Warm	Difference	Cool	Warm	Difference
Distance 1	9.20 (0.56)	4.70 (0.17)	4.50	3.84 (0.08)	3.41 (0.02)	0.43
Distance 2	8.00 (0.51)	4.61 (0.13)	3.39	3.85 (0.07)	3.47 (0.03)	0.38
Distance 3	7.33 (0.42)	4.64 (0.13)	2.69	3.74 (0.06)	3.45 (0.02)	0.29

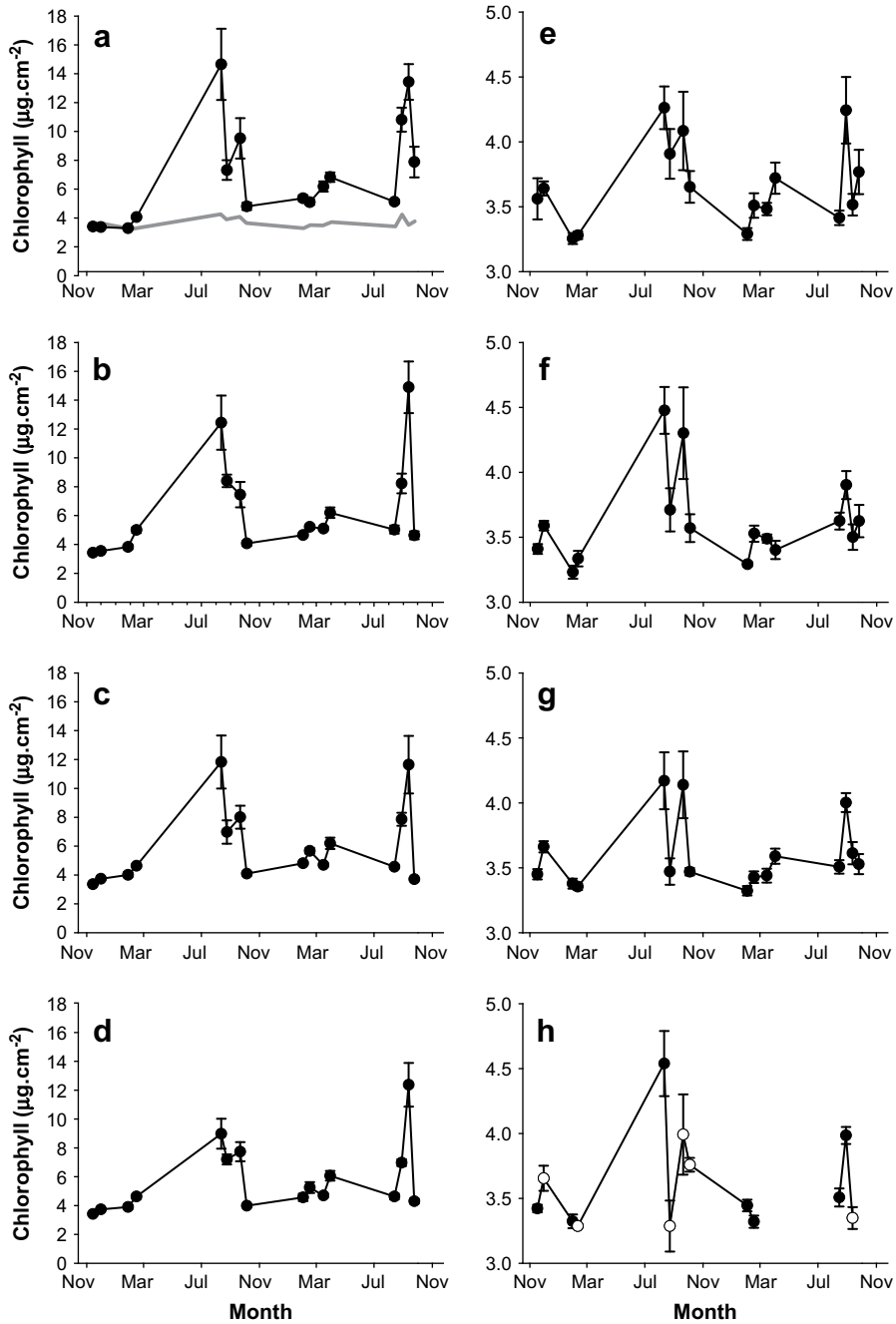

Fig. 1. Amount of chlorophyll (µg cm^{-2}) at 4 distances on the shore: a) Brays Bay distance 1; b) Brays Bay distance 2; c) Brays Bay distance 3; d) Brays Bay distance 4; e) Tambourine Bay distance 1; f) Tambourine Bay distance 2; g) Tambourine Bay distance 3; and h) Tambourine Bay distance 4. Data are averages (±SE) of 3 plots ($n = 12$; ●) or averages of 2 plots ($n = 8$; ○) at each distance. The values from Tambourine Bay at distance 1 (e) are plotted as a grey line in (a) to demonstrate differences in scale on the Y-axis. Estimates of chlorophyll were derived from the spectrometer ratio (see text for explanation).

times sampled in Tambourine Bay was very similar in each of the 2 years (196 in the first year and 193 in the second; Fig. 2). In the first year, it increased to 706 during the cooler months, but increased less (only to 496; Fig. 2) in the cooler months of the second year.

4. Discussion

The majority of studies that have examined seasonal variability of chlorophyll (as an index of MPB biomass) on intertidal mudflats come from European locations. These studies have often shown that amounts of chlorophyll generally increased during the warmer months relative to the cooler months (Cadee and Hegeman, 1977; de Jong and de Jonge, 1995; MacIntyre et al., 1996; Santos et al., 1997; Guarini et al., 1998; Underwood and Kromkamp, 1999; Staats et al., 2001; Migne et al., 2004), although peaks can occur in spring (Admiraal et al., 1982; Sahan et al., 2007), autumn (Davis and McIntire, 1983), or at any other time of the year if local environmental conditions are favourable (Underwood and Kromkamp, 1999). The biological, physical and chemical components of soft-sedimentary habitats interact through a complex network of pathways and feedback loops so when one component of the sediment changes, other components are affected through direct or

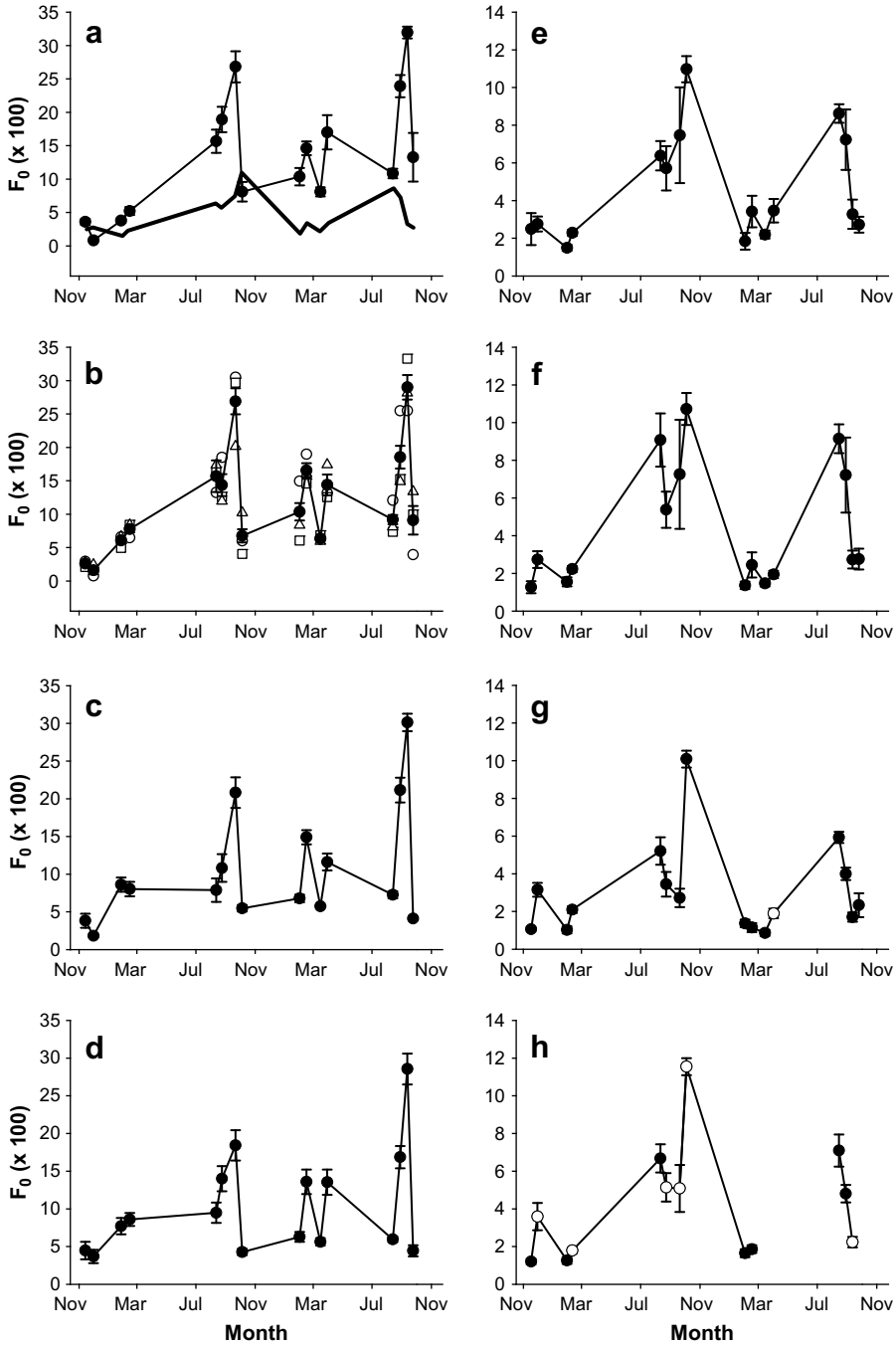

Fig. 2. Biomass (F_0) of algae at 4 distances on the shore: a) Brays Bay distance 1; b) Brays Bay distance 2; c) Brays Bay distance 3; d) Brays Bay distance 4; e) Tambourine Bay distance 1; f) Tambourine Bay distance 2; g) Tambourine Bay distance 3; and h) Tambourine Bay distance 4. Data are averages (±SE) of all 3 plots ($n = 12$; ●) or averages of 2 plots ($n = 8$; ○) at each distance. The values from Tambourine Bay at distance 1 (e) are plotted as a grey line in (a) to demonstrate differences in scale on the Y-axis. The open symbols in b) are data from the individual plots at that distance to illustrate variability in the data.

indirect interactions (Ruddy et al., 1998). It is likely therefore that seasonal patterns in amounts of chlorophyll on NSW intertidal mudflats, like on mudflats elsewhere, are influenced by a combination of "top-down" (e.g. grazing or bioturbation) and "bottom-up" processes (e.g. availability of light and nutrients, temperature, insolation during emersion and supply of MPB from the water column).

This study has shown that mudflats in NSW show strong seasonal patterns in amounts of chlorophyll, with more chlorophyll in the cooler than in the warmer months. This pattern is opposite to that generally found in many northern European estuaries. Latitude and its effects on temperature, insolation and day-length may be an important influencing factor in the timing of seasonal peaks in chlorophyll in European mudflats (Admiraal et al., 1982). At higher European latitudes, where winter temperatures and levels of light are small, amounts of chlorophyll are often greater in spring, summer or autumn, when temperatures are higher and day-length longer than in winter. van Bergeijk et al. (2006) showed that faster growth of some species of diatoms in the Wester- and Oosterschelde estuaries in the Netherlands was correlated with increasing

temperatures. Similar increases in amounts of chlorophyll during the warmer months have been found at higher latitudes in the southern hemisphere. For example, Riaux-Gobin and Bourgoin (2004), in their study of islands in the subAntarctic Indian Ocean (49° S), found that chlorophyll tended to increase during spring and summer. At southern European latitudes, some studies have found no clear seasonal patterns in amounts of chlorophyll (Varela and Penas, 1985; Brotas et al., 1995), which may be due to greater average temperatures and insolation, enabling the continued growth of MPB during the winter months, thus flattening out the seasonal response.

Although NSW and northern Europe largely share the same Koppen-Geiger climate classification (Peel et al., 2007), the amounts of light received by the surface are very different in Sydney (33° S), compared with many northern European locations (~50–55° N). The mudflats studied here are located at lower latitudes, with greater average maximal air temperatures and insolation than in most northern European sites. If growth of MPB was not constrained by minimal winter temperatures and insolation then seasonal variation could be expected to be similar to some southern European locations (i.e. they would have no seasonal pattern). Data from elsewhere do not, however, predict the pattern found here, i.e. more MPB during winter than during summer.

Insolation has been shown to have a significant impact upon growth or survival of algae on mudflats in NSW. Chapman and Tolhurst (2004) showed that there was more chlorophyll on the substratum shaded by the mangrove canopy than on mud exposed to the sun and Murphy et al. (2008a) showed that partial shading of the mud surface from direct sunlight during the warmer months significantly increased amounts of chlorophyll relative to unshaded areas. Total and partial shading of the mud surface from direct sunlight has also been shown experimentally to decrease and increase, respectively, amounts of chlorophyll in experimental plots relative to unshaded sediment (Tolhurst et al., in review). In the present study, amounts of chlorophyll and air temperatures were significantly negatively correlated. It therefore appears that amounts of chlorophyll on emersed mudflats in NSW may be limited by increasing amounts of insolation and air temperature during the warmer months. Thornton and Vissner (2009) also found that benthic chlorophyll-*a* was consistently greater in winter than in summer in salt-marshes in Galveston Island, Texas USA (latitude 29° S), which adds support to the model that increased temperatures and insolation at lower latitudes may limit production of MPB during the warmer months. One study of temporal variations in benthic chlorophyll in an Australian estuary was done in sub-tidal habitats and showed peaks in chlorophyll in all seasons of the year, but most consistently in the austral spring (Lukatelich and McComb, 1986). Unlike intertidal environments, sub-tidal environments are not exposed to desiccation and heat-stress during periods of emersion and this may explain why peaks in chlorophyll can occur at any time of the year in these habitats.

Diatoms and euglenids, which dominate MPB on many European mudflats, can migrate vertically and change their position in the sediment according to the amount of light. This behaviour prevents over-exposure to sunlight (Kromkamp et al., 1998; Underwood et al., 2005) and enables diatoms to avoid damaging levels of UVB (Waring et al., 2007). Filamentous green algae, which were a dominant component of the assemblage on the shores studied here, particularly during winter, are unable to migrate vertically and are therefore exposed to large amounts of direct insolation during summer. These algae may thus be exposed to damaging amounts of insolation during low tide, which may kill the filamentous algae and/or shift the composition of the assemblage to species which are more adapted to high-light conditions, such as diatoms. These processes could thus contribute to the overall decrease in the amounts of chlorophyll in the warmer months. Davis and McIntire (1983) showed that, on a mudflat with large numbers of diatoms in addition to *Enteromorpha* sporelings, patterns of productivity only showed a summer increase when sporelings were present and diatoms alone showed little seasonal pattern.

Although in northern Europe, amounts of chlorophyll may be limited by cooler temperatures and lower levels of insolation during the winter months and changes to temperature and light have been suggested as reasons for the increase in MPB at the end of winter, other factors have also been suggested as important in controlling this pattern, particularly to explain the decrease in MPB at the end of summer. Thus, grazing can play an important role in regulating both the composition and biomass of MPB (e.g. Hagerthey et al., 2002; Sahan et al., 2007); see also reviews by Admiraal (1984), Miller et al. (1996) and Underwood and Kromkamp (1999). Arrival or departure of large numbers of grazers can have a sudden and significant impact upon MPB biomass (Gould and Gallagher, 1990; Sahan et al., 2007). Grazing may similarly play a role in regulating abundance of chlorophyll in NSW mudflats. Intertidal crabs (e.g. *Helocious cordiformis* and *Sesarma erythrodactyla*) are more abundant in mangroves in summer than in winter (Warren, 1987) and appear to show a similar pattern on these mudflats (pers. obs.), seldom emerging from burrows during the colder months. These crabs can process large amounts of mud when feeding, thus potentially reducing MPB by direct consumption and by disturbance of the sediment surface, increasing the probability of re-suspension when the tide comes in (Reinsel, 2004).

On the mudflats studied here, there are no other grazers that reach large abundances equivalent to those reported elsewhere to reduce levels of MPB as a direct result of grazing pressure, e.g. *Corophium volutator* in The Netherlands (de Jong and de Jonge, 1995) or *Hydrobia ulvae* in the Danish Wadden Sea (Austen et al., 1999). In Europe, Hydrobids, for example, can reach densities of 10,000 m^{-2} (Barnes, 2001), but benthic assemblages on the intertidal mudflats in Sydney harbour are dominated by polychaetes and oligochaetes (e.g. Chapman and Tolhurst, 2004) and the most abundant snails, Salinator spp., seldom reach densities of more than a few hundred per square metre (pers. obs.). Therefore, although the relative importance of grazing in determining seasonal patterns in NSW is presently unknown, it may be less important than the direct effects of increased summer temperature and insolation on algal growth or survival (Murphy et al., 2008a; Tolhurst et al., in review). Chapman and Tolhurst (2004) found, for example, no consistent correlations between invertebrate assemblages and chlorophyll at any of the spatial scales measured, suggesting no clear dependence of these fauna on amounts of MPB (see also Santos et al., 1996).

Because of the large number of fine strands of macro-algae in the MPB at some times of the year, which is not reported for the MPB in many mudflats in northern Europe, the response of infauna and other biogeochemical components of the sediment to variability in MPB is also likely to be different from those reported from other studies. This raises several important questions about how seasonal variability in MPB might affect the structure and functioning of these different benthic assemblages. In northern Europe, for example, increased energetic requirements of infauna occur in spring and summer (Hubas et al., 2006), which is coincident with seasonal increases in micro-algal food. In NSW, energetic requirements of infauna are also increased in summer at a time when food-resources may be reduced by increasing temperatures and insolation. Alternatively, the active grazers in summer may reduce the standing stock of MPB, as they can on adjacent rocky shores (Underwood, 1984a). A small standing stock of MPB may still produce adequate biomass to sustain grazers (Pinckney et al., 2003). The relative

importance of abiotic factors and grazers in controlling biomass of chlorophyll on these intertidal mudflats requires further manipulative experimental studies (Underwood, 1985), as done for adjacent rocky shores where a series of experiments clearly identified the relative roles or grazers and abiotic factors in controlling patterns of micro-algae (Underwood, 1984b).

Sudden and sometimes large changes in amounts of chlorophyll appear to be common features of European (e.g. Santos et al., 1997; Kornman and De Deckere, 1998; Underwood and Kromkamp, 1999) and NSW mudflats (this study). Superimposed on variation at the scale of seasons are large changes in amounts of chlorophyll that occur from week to week. Such changes were found in the warmer and cooler months, but the greatest weekly variations occurred only in the cooler months. The causes of these fluctuations cannot be determined without further experiments, but they may have something to do with episodes of warm temperatures and increased insolation that can occur in NSW during the cooler months. The large amount of week-to-week variability has significant implications for the design of sampling strategies for measuring seasonal variation (as described in Underwood, 1994). If only one sampling time had been used to represent seasonal change, different conclusions could have been reached, depending on the week that happened to be sampled in each season. For example, the amount of chlorophyll at the 8th time of sampling (i.e. during a 'cool' season; point 8 in Figs. 1a–d and 2a–d) is much less than the other sampling times for this season. If only this time had been sampled, it would have been concluded that, during the first year of sampling at Tambourine Bay, amounts of chlorophyll were not significantly greater in the cool than in the warm season. Such problems in sampling have been identified by other researchers (e.g. Rizzo and Wetzel, 1985; Santos et al., 1997) and underscore the need for properly replicated sampling in studies of spatial or temporal variability of MPB. To identify seasonal variation unambiguously, it is not only necessary to replicate seasons (i.e. sample over at least 2 years), but also to have adequate temporal replication within each season to separate seasonal change from other short-term patterns of variability.

This study also found more chlorophyll and larger seasonal variability on the upper shore (Distance 1) than on the lower shore (Distance 3; Table 3), particularly at Brays Bay. Similar patterns have been reported for European or North American mudflats (Davis and McIntire, 1983; de Jong and de Jonge, 1995; Staats et al., 2001), although de Jong and de Jonge (1995) showed considerable variability among different transects. This pattern may be due to a combination of factors, including increased re-suspension by wave action (Demers et al., 1987; de Jong and de Jonge, 1995) and/or the shorter emersion times allowing less time for migratory MPB to stay on the surface during periods of low tide. de Brouwer et al. (2000) also showed that the upper shore showed the largest seasonal differences for the Biezelingse Ham mudflat in the Westerschelde (Netherlands).

Although there were large differences in amounts of MPB between the 2 bays studied here, the temporal patterns were generally consistent, although Tambourine Bay is in an urbanised area and Brays Bay is adjacent to an area which was industrialized for many years, but is now largely bordered by parkland. Sediments at Brays Bay are finer-grained than are sediments at Tambourine Bay (unpublished data), which may contribute to the differences in chlorophyll between these sites because fine sediments are often associated with greater amounts of chlorophyll (e.g. Cammen, 1982; de Jong and de Jonge, 1995). Staats et al. (2001) showed that the large summer values of chlorophyll were only found in parts of the shore with greater amounts of mud. Here, too, patterns were much stronger on the shore with apparently muddier sediment.

Other properties of the sediment, such as grain-size, total and colloidal carbohydrate, may also change over similar timescales in response to changes in amounts of chlorophyll (de Brouwer et al., 2000; Staats et al., 2001). These may, in turn, determine the suitability of the sediment as a habitat for benthic fauna (reviewed by Gray, 1974; Snelgrove and Butman, 1994). Sediment stability has been linked to amounts of chlorophyll in the upper layers of the sediment in NSW (Murphy et al., 2008a), which supports the pattern reported for many European estuaries (reviewed by Miller et al., 1996). Increases in sediment stability in European mudflats are mainly the consequence of increased diatom biomass (de Brouwer et al., 2005; Tolhurst et al., 2008), but, in NSW, the increases in green filamentous algae and diatoms in winter may enhance sediment stability. It must be remembered, however, that at the same time, there is a decrease in the numbers of foraging crabs, which is also likely to reduce disturbance of the sediment surface and increase stability.

The patterns measured by the PAM and the field spectrometer were largely consistent. At some times, however, there were differences. For example, at Brays Bay, on 12/08/06, 24/08/06 and 21/09/06, there was a general decrease in amounts of chlorophyll derived from the reflectance ratio (Fig. 1a–d), but an increase in F_0 (Fig. 2a–d). The spectrometer ratio and F_0 measure different things – the former chlorophyll and the latter any substance in the mud that fluoresces when excited by the wavelengths of light used by the PAM (655 nm). Vertical migration by diatoms or euglenids between the measurement of reflectance and that of F_0 could also cause differences between the measurements. Because the sediment was dark-adapted for 15 min prior to the measurement of F_0, vertical upward migration by some MPB may have caused a temporary increase in the amounts of MPB at the surface and, hence, an increase in the F_0 measurement. This would not happen for the Spectrometer readings because the surface was not dark-adapted. This is, however, unlikely to explain the differences observed, however, because they were very large. These differences are more likely to have been caused by the PAM detecting fluorescence of organic materials in the mud other than active chlorophyll, e.g. detritus flushing out from adjacent mangrove forests. To test the hypothesis that changes in the compositional makeup of the assemblage were contributing to the differences between F_0 and the chlorophyll derived from the reflectance ratio, absorption by pigments other than chlorophyll-*a* was quantified from the reflectance spectra, but there were no significant differences in the pigment composition over the period when the two measures of chlorophyll differed. Other studies have described similar differences between these different means of measuring of chlorophyll (e.g. see Fig. 9 in Kromkamp et al., 2006).

It is difficult to make progress on understanding ecological processes without first understanding ecological patterns (Underwood et al., 2000). Knowledge of spatial or temporal patterns of MPB is an important step towards understanding the processes that influence them, but does not, in itself, identify the relative importance of the many underlying ecological processes which cause these patterns. This paper has provided the first quantitative description of seasonal patterns of chlorophyll on intertidal mudflats in NSW, showing that the seasonal patterns of abundance found in previous studies, mainly in Europe, are not universal. Yet much of our understanding of soft-sediment ecology comes from studies in Europe and North America and many of these studies have not separated seasonal variation from other temporal patterns because of poor temporal replication. We therefore demonstrate the need to quantify patterns across many different habitats before simply transferring paradigms or ideas about the relative importance of different processes from one environment to another

(Underwood and Petraitis, 1993). It is clear that the seasonal processes influencing MPB abundance on mudflats in many parts of the world do not apply to mudflats around Sydney, NSW, Australia. Manipulative experiments are now needed to unravel the complex interplay of ecological processes that cause this variation and how these operate across a cascade of spatial and temporal scales.

5. Conclusions

Amounts of chlorophyll on mudflats in New South Wales were greater in the cooler than in the warmer months. This is opposite to what has been observed for many European mudflats, where peaks in amounts of chlorophyll have been often found to occur in spring, summer or autumn. Latitude and its effects on temperature, insolation and day-length may play an important role in regulating biomass of MPB. At higher latitudes, cooler temperatures, low levels of light and shorter days may limit the growth of MPB (and therefore amounts of chlorophyll) in the cooler months. At lower latitudes, as in the present study, increased insolation and temperatures during the warmer months may limit the growth of MPB in the warmer months. Amounts of chlorophyll on mudflats were negatively correlated with chlorophyll in the mudflats studied here.

Amounts of chlorophyll were also greater on the upper than the lower shore, consistent with patterns found on mudflats in Northern European locations.

Large differences in the amount of chlorophyll were found between the 2 bays sampled, but seasonal patterns of abundance were consistent. Large variations were found in both bays over relatively short (week to week) timescales, underscoring the need for properly replicated sampling in studies of seasonal variation.

Further experiments are required to unravel effects of insolation and temperature from the effects of grazing animals, such as crabs, which become more active in the warmer months.

Differences in the amounts of chlorophyll detected by PAM and field spectrometry were not caused by changes in the composition of the algal assemblage.

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